CS683 Programming Assignment 2
Task 2 Summary: L2 Cache Replacement Policies

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1 I-RRIP Policy (Task 2A)

1.1 Insertion and Promotion

An instruction miss can delay fetching more instructions, so I-RRIP gives instruction lines a small advantage. Each line has an RRPV from 0 to 3; a lower value means it is less likely to be evicted. Instruction fills start at 1, while data fills stay at the RRIP value of 2. This gives instruction lines an edge without pushing data lines far back. A hit moves an instruction line to 0 and a data line to 1. Both get promoted, but the instruction line stays ahead. An instruction request is a LOAD or PREFETCH marked as an instruction; translations count as data requests.

1.2 Victim Selection

The policy uses an invalid way first. If every way is full, it evicts a data line at RRPV 3 when possible. At the same RRPV, this keeps an instruction line in the cache longer. If there is no data line ready, the policy can evict any line at 3, so instruction lines cannot block new data forever. When no line is ready, it raises all RRPVs and checks again.

1.3 Results

Trace	Policy	IPC	Speedup	L2 MPKI	Instr. MPKI	Data MPKI
Trace 1	LRU	0.232345	1.000000	17.695640	10.337680	7.357960
Trace 1	SRRIP	0.232231	0.999510	17.638680	10.561800	7.076880
Trace 1	I-RRIP	0.232191	0.999338	17.598600	9.883480	7.715120
Trace 2	LRU	0.251158	1.000000	27.356799	18.853279	8.503520
Trace 2	SRRIP	0.251806	1.002579	25.745799	17.781279	7.964520
Trace 2	I-RRIP	0.256618	1.021741	22.129439	12.974319	9.155120
Trace 3	LRU	0.382065	1.000000	3.954240	1.124080	2.830160
Trace 3	SRRIP	0.381739	0.999149	4.039480	1.197040	2.842440
Trace 3	I-RRIP	0.381977	0.999772	3.896360	1.022120	2.874240

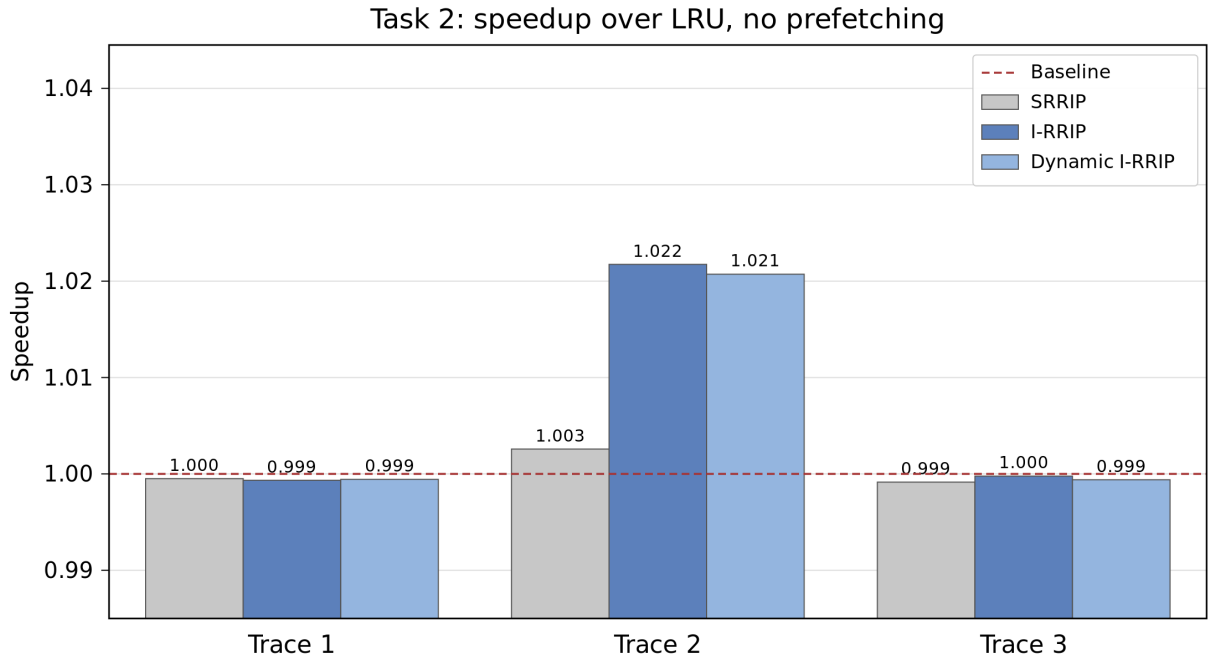


Figure 1: Speedup over LRU for each trace. Dynamic I-RRIP uses the selected 15% threshold.

On Trace 2, I-RRIP is 2.17% faster than LRU. SRRIP is 0.26% faster. I-RRIP is slightly slower than LRU on Trace 1 and Trace 3.

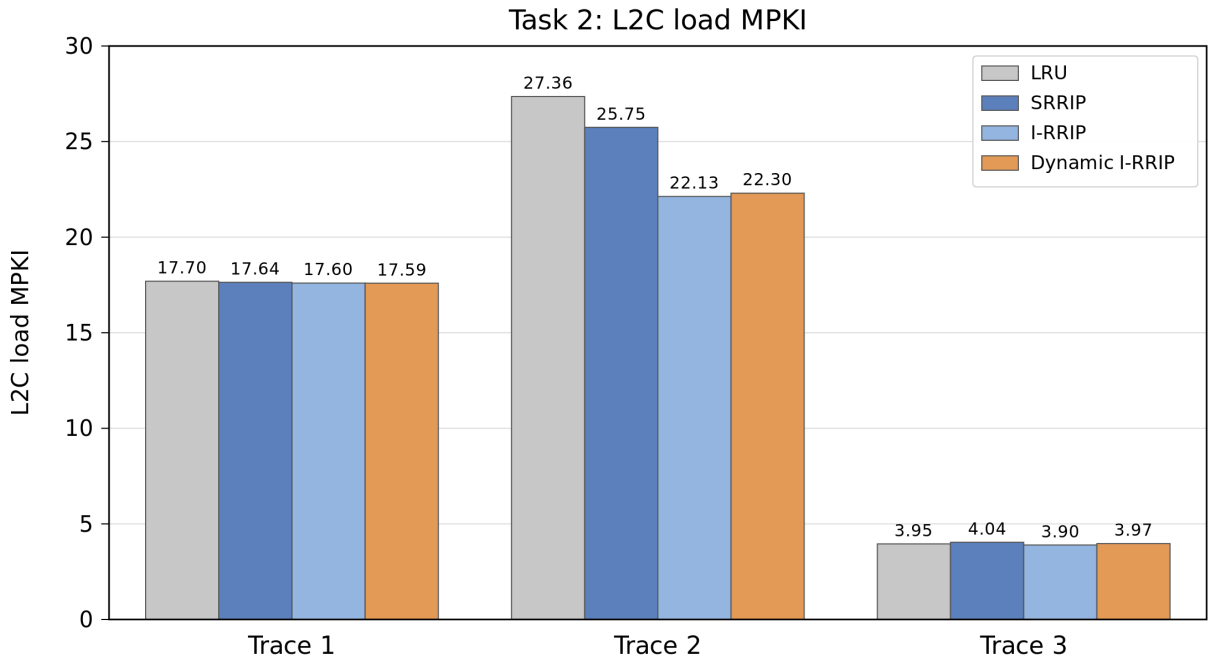


Figure 2: Total L2 load MPKI for each policy and trace.

On Trace 2, I-RRIP lowers total L2 load MPKI from 27.36 to 22.13. The change on Trace 1 is small, from 17.70 to 17.60. Trace 3 has fewer load misses overall, so the differences between policies are smaller there.

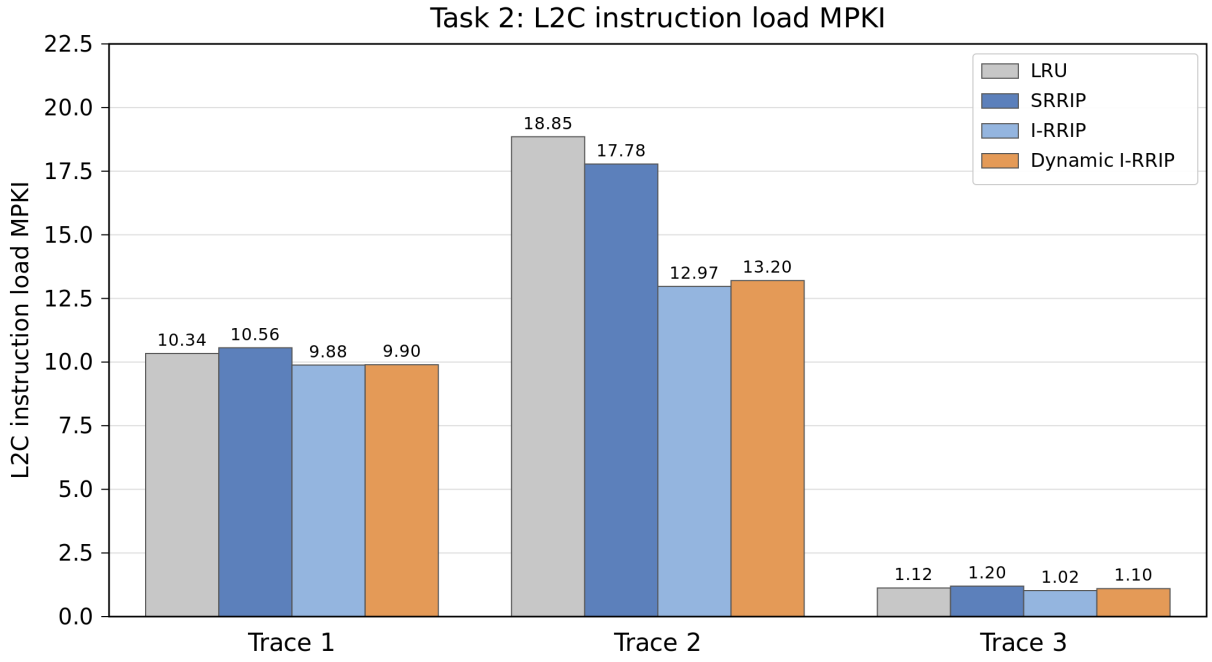


Figure 3: Instruction load MPKI at L2.

I-RRIP lowers instruction MPKI on all three traces. Compared with LRU, it falls by 4.39% on Trace 1,

31.18% on Trace 2, and 9.07% on Trace 3. The largest change is on Trace 2, from 18.85 to 12.97.

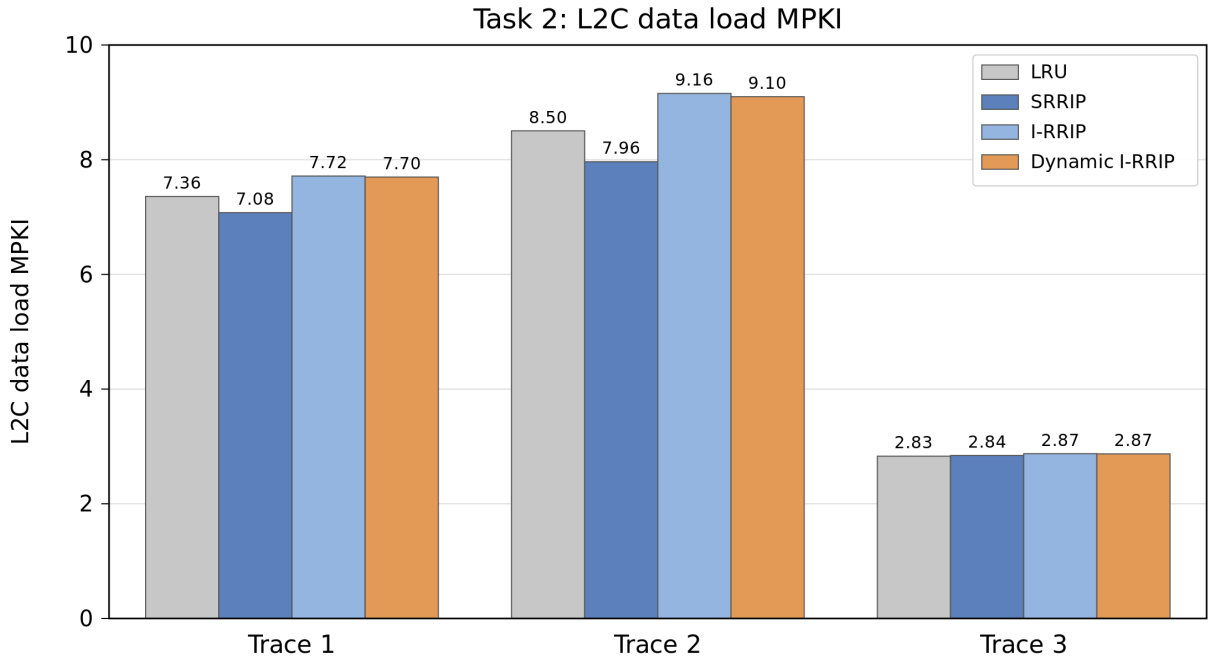


Figure 4: Data load MPKI at L2.

I-RRIP also causes more data misses: data MPKI rises by 4.85% on Trace 1, 7.66% on Trace 2, and 1.56% on Trace 3. On Trace 2, the drop in instruction misses is large enough to improve IPC. On the other two traces, lower instruction MPKI does not lead to a speedup.

2 Dynamic I-RRIP Policy (Task 2B)

2.1 Leader Sets

Leader sets always use RRIP and track instruction and data requests. The remaining sets use I-RRIP or RRIP based on those counts. This way, instruction priority is used only when the leader sets show it may help. Sets 0, 32, 64, ..., 992 are leaders. That samples 32 of the 1,024 sets (3.125%) and leaves 992 sets to follow the chosen policy. Even spacing samples the cache while leaving most sets free to follow the selected policy. We kept the number of leaders fixed during testing.

2.2 Switching Decision

The leaders keep four counts: instruction accesses and misses, plus data accesses and misses. Each hit or fill counts as an access; a fill also counts as a miss. After 1,024 leader accesses, the policy calculates this share:

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Instruction share = (instruction accesses + instruction misses)
                          / (total accesses + total misses)
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If the share is above the threshold, the other sets use I-RRIP; otherwise they use RRIP. The formula counts both accesses and misses, so a miss has twice the weight of a hit. This makes instruction priority

depend on the instruction share and its misses, instead of only on how often instructions are accessed. The counts reset after each decision. Follower sets start in RRIP, and changing modes leaves their RRPVs as they are.

2.3 Threshold Sensitivity

We used a 1,024-access window so a few requests would not change the mode, but the policy could still respond during a run. We kept the window fixed and tested five thresholds: 15%, 20%, 25%, 30%, and 35%. Speedup over LRU at each threshold:

Threshold	Trace 1	Trace 2	Trace 3	Geo. mean	Change
15%	0.999434	1.020720	0.999394	1.006466	+0.6466%
20%	0.999470	1.019846	0.999128	1.006101	+0.6101%
25%	0.999993	1.019846	0.999128	1.006277	+0.6277%
30%	1.000338	1.019579	0.999160	1.006315	+0.6315%
35%	1.000271	1.019579	0.999073	1.006264	+0.6264%

The 15% threshold has the best geometric mean speedup: 1.006466, or 0.65% above LRU. We used it for the main comparison. The scores are close, and we tested the thresholds on these three traces only.

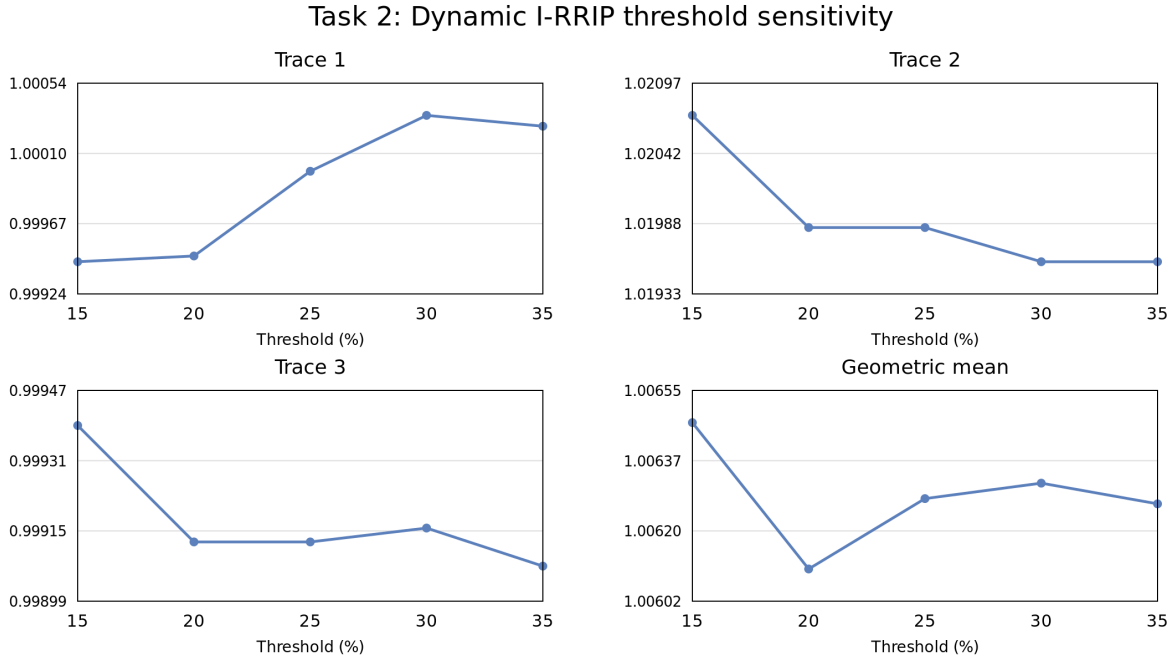


Figure 5: Speedup over LRU at five thresholds; each panel uses its own vertical scale.

2.4 Results at the Selected Threshold

Dynamic I-RRIP results at 15%:

Trace	Policy	IPC	Speedup	L2 MPKI	Instr. MPKI	Data MPKI
Trace 1	Dynamic	0.232213	0.999434	17.593280	9.896080	7.697200
Trace 2	Dynamic	0.256362	1.020720	22.300759	13.202439	9.098320
Trace 3	Dynamic	0.381833	0.999394	3.969520	1.099320	2.870200

At 15%, Dynamic I-RRIP improves Trace 2 IPC by 2.07%. It is 0.06% below LRU on both Trace 1 and Trace 3. Its geometric mean speedup is close to static I-RRIP, but is slightly lower.

2.5 Results Across All Thresholds

Dynamic I-RRIP results for all five thresholds:

Trace	Threshold	IPC	Speedup	L2 MPKI	Instr. MPKI	Data MPKI
Trace 1	15%	0.232213	0.999434	17.593280	9.896080	7.697200
Trace 1	20%	0.232222	0.999470	17.602560	9.912320	7.690240
Trace 1	25%	0.232343	0.999993	17.573240	9.934680	7.638560
Trace 1	30%	0.232423	1.000338	17.517720	9.959520	7.558200
Trace 1	35%	0.232408	1.000271	17.456960	10.140920	7.316040
Trace 2	15%	0.256362	1.020720	22.300759	13.202439	9.098320
Trace 2	20%	0.256142	1.019846	22.474479	13.379399	9.095080
Trace 2	25%	0.256142	1.019846	22.474479	13.379399	9.095080
Trace 2	30%	0.256075	1.019579	22.514759	13.431079	9.083680
Trace 2	35%	0.256075	1.019579	22.514759	13.431079	9.083680
Trace 3	15%	0.381833	0.999394	3.969520	1.099320	2.870200
Trace 3	20%	0.381731	0.999128	4.032960	1.160040	2.872920
Trace 3	25%	0.381731	0.999128	4.032960	1.160040	2.872920
Trace 3	30%	0.381744	0.999160	4.030680	1.159320	2.871360
Trace 3	35%	0.381710	0.999073	4.031480	1.159160	2.872320

A higher threshold makes it harder to switch to I-RRIP, so the policy stays in RRIP more often. Some thresholds produce the same results on Trace 2: 20% and 25%, and 30% and 35%.

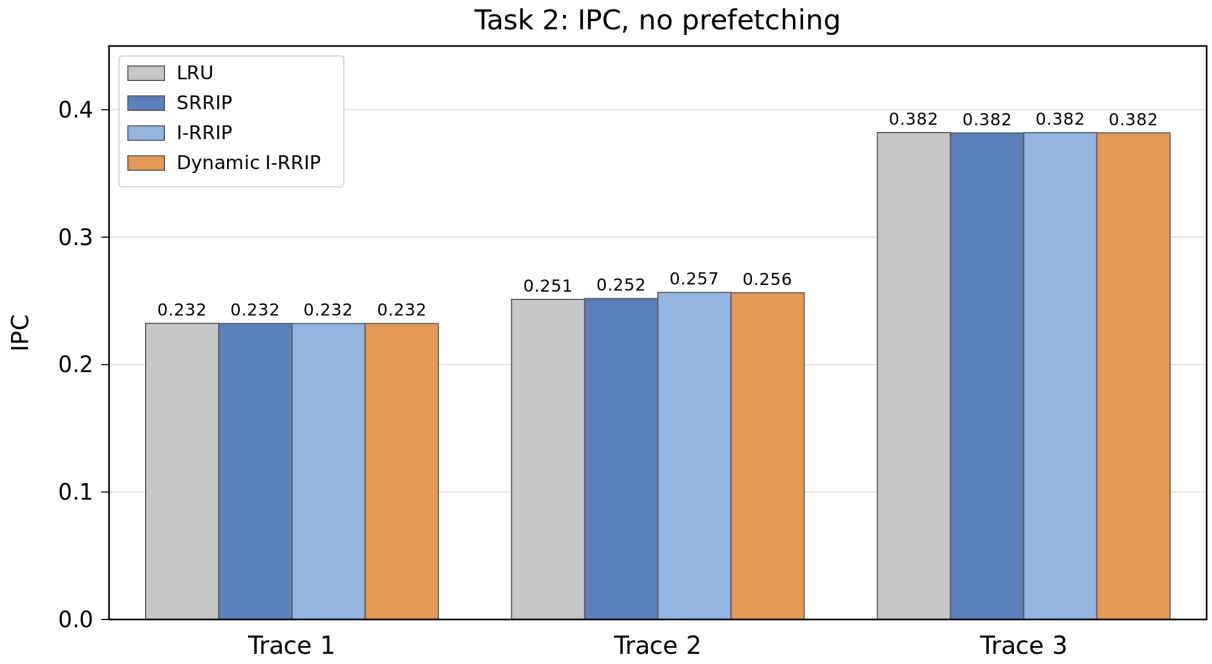


Figure 6: IPC for the four policies; Dynamic I-RRIP uses threshold 15%.

Trace 2 has the largest IPC gain. Trace 3 has the highest IPC, but the replacement policy makes little difference there.

3 Summary

I-RRIP lowers instruction load MPKI on all three traces, but raises data load MPKI. It improves IPC on Trace 2 and is slightly slower than LRU on Trace 1 and Trace 3. For Dynamic I-RRIP, the best tested threshold was 15%, with a geometric mean speedup of 1.006466. Static I-RRIP reached 1.006896, and SRRIP reached 1.000412. Dynamic I-RRIP helps on Trace 2, but not on every trace.